

Capstone Section 3 - MongoDB

Commands below were run against a local MongoDB 7.0.14 instance via mongosh. Output is captured directly from the server, not simulated.

Create a new entry in the database

```
db = db.getSiblingDB("lms");

db.students.insertOne({
  name: "Cara Ong",
  email: "cara.ong@example.com",
  enrolledCourses: ["AI & Machine Learning"],
});

db.students.find();
```

Result:

```
Insert result:
{
  acknowledged: true,
  insertedId: ObjectId('6ab8908e85d6609cb0be0e68')
}

db.students.find():
{
  _id: ObjectId('6ab8908e85d6609cb0be0e68'),
  name: 'Cara Ong',
  email: 'cara.ong@example.com',
  enrolledCourses: [
    'AI & Machine Learning'
  ]
}
```

RESTful API on top of this data (Express + Mongoose)

express-app/server.js exposes the same students collection through a small REST API:

- GET /students - list all students
- POST /students - add a new student (validates name/email are present)

Verified with curl against the running server:

```
$ curl -X POST -H "Content-Type: application/json" \
      -d '{"name":"Ben Lim","email":"ben@example.com","enrolledCourses":["Full Stack Web Development"]}' \
      http://localhost:4000/students

{"name":"Ben Lim","email":"ben@example.com","enrolledCourses":["Full Stack Web Development"], "_id":"6ab890afe1ffb776fa23aa74", "__v":0}

$ curl http://localhost:4000/students
```

```
[
    {
        "_id": "6ab8908e85d6609cb0be0e68",
        "name": "Cara Ong",
        "email": "cara.ong@example.com",
        "enrolledCourses": ["AI & Machine Learning"]
    },
    {
        "_id": "6ab890afe1ffb776fa23aa74",
        "name": "Ben Lim",
        "email": "ben@example.com",
        "enrolledCourses": ["Full Stack Web Development"],
        "__v": 0
    }
]
```